

- 7 (a) waves from the double slit are coherent/constant phase difference B1
- waves (from each slit) overlap/superpose/meet (**not** interfere) B1
- maximum/bright fringe where path difference is $n\lambda$
 or phase difference is $n360^\circ/2\pi n$ rad
- or minimum/dark fringe where path difference is $(n + \frac{1}{2})\lambda$
 or phase difference is $(2n + 1) 180^\circ/(2n + 1)\pi$ rad B1 [3]
- (b) $v = f\lambda$ C1
 $\lambda = (3 \times 10^8) / 670 \times 10^{12} = 448$ (or 450) (nm) M1 [2]
- (c) $w = 12 / 9$ C1
 $a (= D\lambda/w) = (2.8 \times 450 \times 10^{-9}) / (12 / 9 \times 10^{-3})$ [allow nm, mm] C1
 $= 9.5 \times 10^{-4} \text{ m}$ [$9.4 \times 10^{-4} \text{ m}$ using $\lambda = 448 \text{ nm}$] A1 [3]
- (d) (red light has) larger/higher/longer wavelength (must be comparison) M1
 fringes further apart/larger separation A1 [2]